

# **Frontend Progress Report**

## **AI Expense Analyzer**

**Team CSE4204-8A-T04**

Section: 8A

Team Leader: Md. Shakinul Islam

*Submitted to: Md. Riaz Mahmud, Assistant Professor,  
Dept. of CSE, NUBTK*

Week 07 Submission — Frontend Development &  
Backend Integration

July 6, 2026

## 1. Project Title

AI Expense Analyzer — an AI-powered personal finance management application that helps users track income and expenses, visualize spending patterns, and receive AI-generated budgeting advice through the Google Gemini API.

## 2. Team Information

| Serial | Name                         | Student ID  | Technology           | Responsibility                   |
|--------|------------------------------|-------------|----------------------|----------------------------------|
| 1      | Shakinul Islam (Team Leader) | 11220120822 | MongoDB              | Database Management & Testing    |
| 2      | Naim Sheikh                  | 11210320634 | Flutter              | Frontend Development & UI Design |
| 3      | Ayon Banerjee                | 11220120820 | Node.js + Express.js | Backend Development              |
| 4      | Afia Rahman                  | 11220320870 | Gemini API           | AI Integration                   |

## 3. Frontend Technology Stack

- Framework: Flutter (Dart) — cross-platform mobile UI, single codebase for Android and iOS
- State & Navigation: Flutter routing with named routes and protected route guards
- HTTP Client: Dio / http package for REST API communication with the Node.js + Express.js backend
- Local Session Storage: Secure storage for the JWT access token issued at login
- UI Components: Material-based custom widgets — gradient app bars, card-based dashboards, bottom navigation bar (Dashboard, Add, AI, Profile)
- Charts & Visuals: Native chart widgets for income/expense summaries on the dashboard

## 4. Completed Pages

The following major screens have been implemented and closely follow the approved UI/UX design from Week 05:

| Page                     | Status    | Notes                                                                                |
|--------------------------|-----------|--------------------------------------------------------------------------------------|
| Landing / Welcome Screen | Completed | App branding, Login and Create Account entry points                                  |
| Login Page               | Completed | Email + password fields, link to Sign Up                                             |
| Registration Page        | Completed | Full name, email, password, and confirm password fields                              |
| Dashboard                | Completed | Balance, income, expense summary cards; date-range filters; recent transactions list |
| Add Transaction Page     | Completed | Income/Expense toggle, amount input, category selection grid                         |

| Page                | Status         | Notes                                                                                        |
|---------------------|----------------|----------------------------------------------------------------------------------------------|
| User Profile Page   | Completed      | Profile summary, income/expense/transaction/savings-goal cards, personal information section |
| AI Assistant Page   | Completed      | Chat-style interface, "Generate Insight" action, conversation history                        |
| Notifications Panel | In Progress    | Bottom-sheet notification list; currently displays empty state                               |
| Admin Panel         | Not Applicable | Not required for this project's scope                                                        |

## 5. Backend APIs Integrated

The frontend has been connected to the following backend endpoints exposed by the Node.js + Express.js server, replacing static/hardcoded data with live data from MongoDB:

| Endpoint           | Method     | Integrated In                                 |
|--------------------|------------|-----------------------------------------------|
| /api/auth/register | POST       | Registration Page                             |
| /api/auth/login    | POST       | Login Page                                    |
| /api/auth/logout   | POST       | Profile Page (logout action)                  |
| /api/user/profile  | GET / PUT  | Profile Page                                  |
| /api/transactions  | POST / GET | Add Transaction Page, Dashboard               |
| /api/dashboard     | GET        | Dashboard summary cards & recent transactions |
| /api/ai/report     | POST       | AI Assistant Page ("Generate Insight")        |
| /api/notifications | GET        | Notifications Panel                           |

## 6. Authentication Flow

- User registers via the Create Account page; credentials are sent to /api/auth/register and the password is hashed with bcrypt on the server.
- On login, credentials are verified and a signed JWT is returned and stored securely on the device.
- The JWT is attached as a Bearer token to all subsequent protected API calls (profile, transactions, dashboard, AI reports, notifications).
- Protected routes (Dashboard, Add Transaction, Profile, AI Assistant) are only reachable once a valid token is present; unauthenticated access redirects to the Login page.
- Logout clears the stored token and returns the user to the Welcome/Login screen.

## 7. Current Development Progress

Overall frontend development is substantially complete for this milestone. All primary screens listed in the Week 06 UI/UX design are implemented and wired to live backend data rather than static placeholders.

- Core navigation (Dashboard → Add → AI → Profile) is fully functional via the bottom navigation bar.
- Dashboard correctly reflects real transaction totals (e.g., balance, income, and expense figures update after a new transaction is added).
- The AI Assistant screen successfully calls the AI insight endpoint and renders formatted expense-analysis text with money-saving suggestions.
- Remaining work: populate the Notifications panel with live budget-alert data, add form-level validation messages across all input fields, and complete responsive testing across tablet screen sizes.

## 8. GitHub Repository Link

<https://github.com/shakinul-islam/AI-Expense-Analyzer>

## 9. Screenshots

Screenshots below were captured directly from the running application on July 6, 2026.

### 9.1 Landing / Welcome Screen

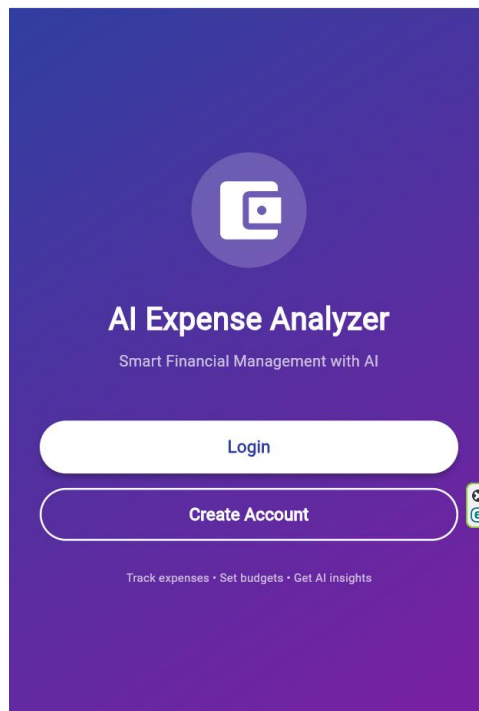


Figure 1: Welcome screen with Login and Create Account options

## 9.2 Login Page

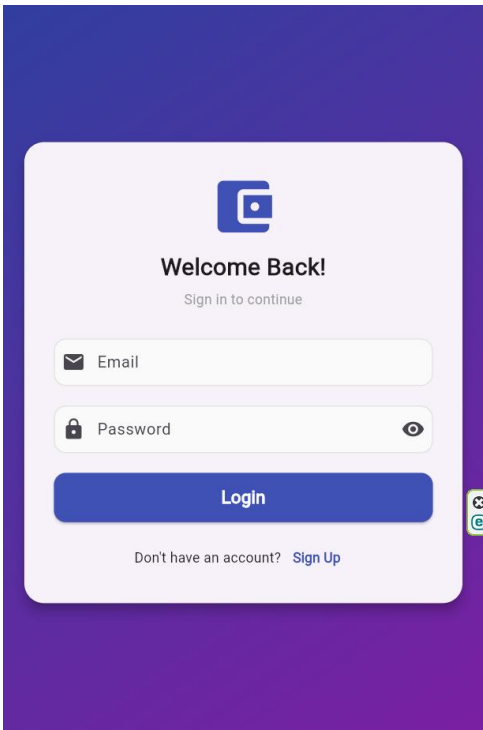


Figure 2: Login page with email and password fields

## 9.3 Registration Page

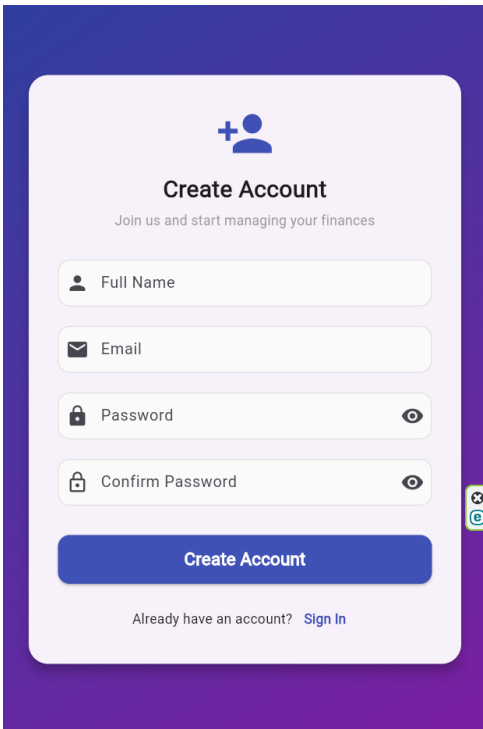


Figure 3: Create Account page with full registration form

9.4 Dashboard

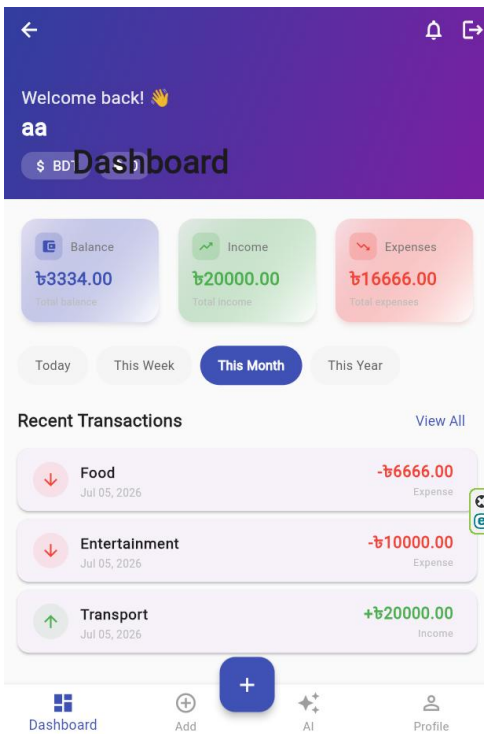


Figure 4: Dashboard showing balance, income, expenses, and recent transactions

9.5 User Profile Page

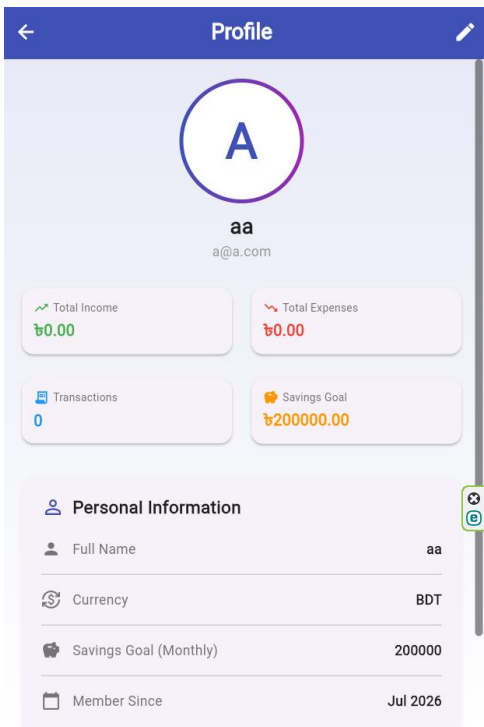


Figure 5: Profile page with account summary and personal information

9.6 Add Transaction Page

Add Transaction

Add New Transaction

Enter the details of your transaction below

Transaction Type

Expense

Income

₹ Amount

Category

Food

Transport

Shopping

Entertainment

Bills

Health

Figure 6: Add Transaction form with type toggle and category grid

9.7 AI Assistant Page

AI Assistant

Generate Insight

Chat cleared. Start a new conversation!

Just now

Ask about your finances...

Figure 7: AI Assistant chat interface (initial state)

Page 7

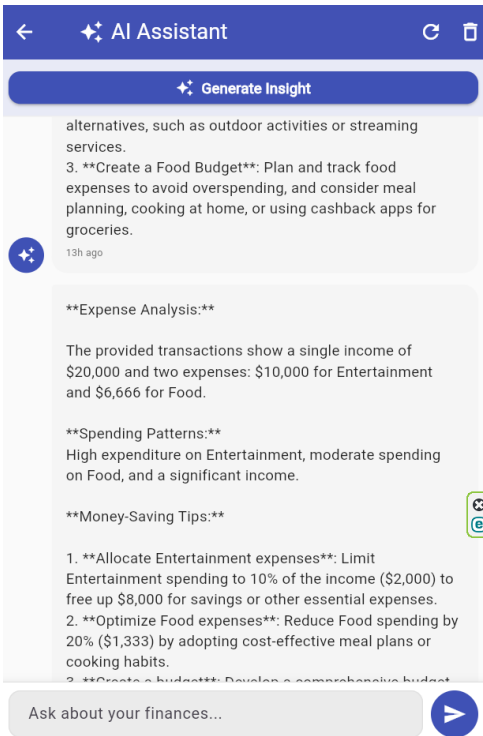


Figure 8: AI-generated expense analysis and money-saving tips

9.8 Notifications Panel

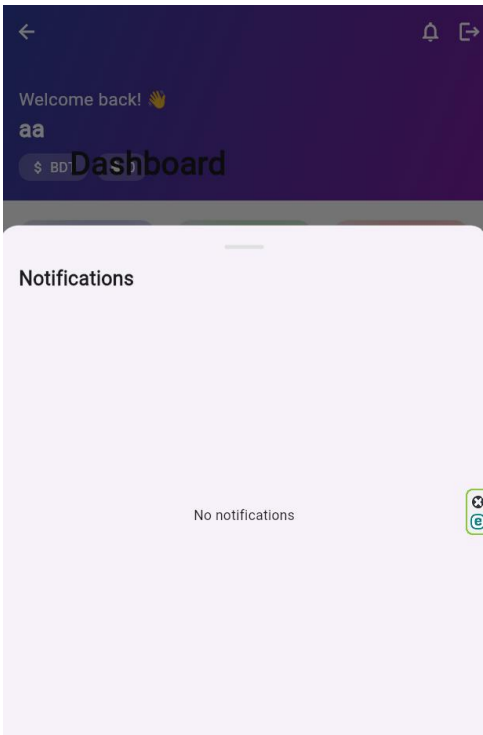


Figure 9: Notifications panel (currently empty state, backend integration in progress)